The Tech Domino Effect: Volatility Transmission from Mega-Caps to the S&P 500

Vo Thi Quynh Tram, Lai Chiu Lok, Huei-Wen Teng

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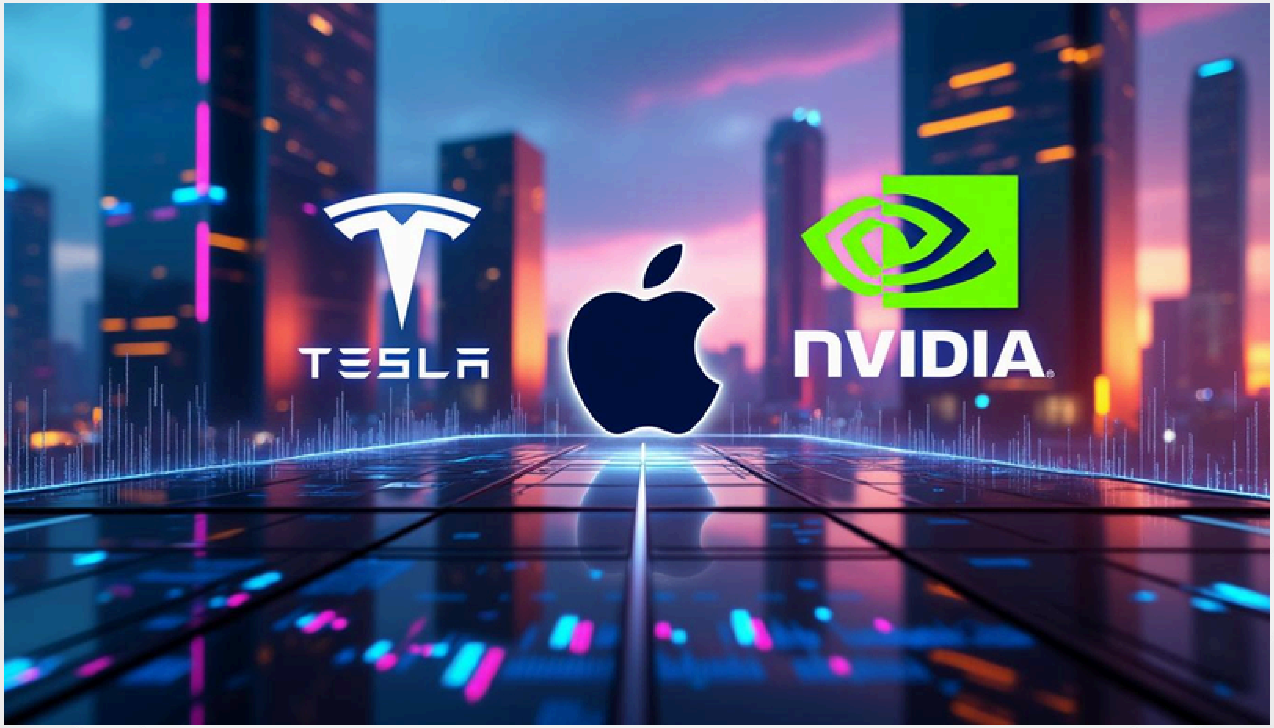
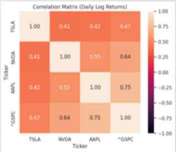
Motivation

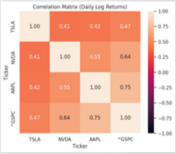
- Rising dominance of mega-cap tech firms in the market.
- Potential volatility spillovers affecting overall market movements.
- Need to assess systemic risk from concentrated market influence.





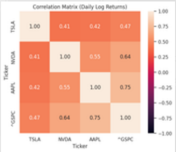
Outline

1. Motivation
2. Data
3. Linear Regression
4. Results
5. Conclusion



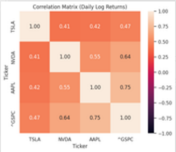
Data Preprocess_Data Overview

Source	Yahoo Finance (Adjusted Close)
Period	2015–2024 (Daily)
Target Assets	Apple (AAPL), Nvidia (NVDA), Tesla (TSLA), S&P 500 (Benchmark)
Transformation	Log returns → Rolling Volatility (annualized)



Data Preprocess_Variables

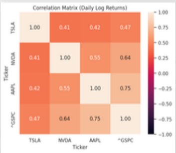
Variable	Definition	Role
RV_SPX	Rolling Volatility of S&P 500	Dependent
RET_AAPL, RET_NVDA, RET_TSLA	Log Returns of tech firms	Independent
RV_AAPL, RV_NVDA, RV_TSLA	Rolling Vol of tech firms	Spillover Drivers



Data Preprocess_Descriptive Statistics

Ticker	count	mean	std	min	25%	50%	75%	max	Skew	Kurtosis	JB_pvalue
AAPL	2514	0.00093	0.017934	-0.137708	-0.007355	0.001	0.010093	0.113158	-0.201228	5.457399	0
NVDA	2514	0.002248	0.030378	-0.207712	-0.012514	0.002648	0.017472	0.260876	0.208734	6.829886	0
TSLA	2514	0.001333	0.035911	-0.236518	-0.016292	0.001264	0.019073	0.198187	-0.066409	4.410539	0
SPX	2514	0.000419	0.011272	-0.127652	-0.00378	0.000639	0.00574	0.089683	-0.81083	15.750819	0

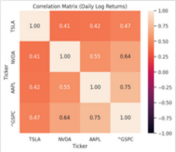
- **All assets: high variance (Std highest for TSLA)**
- **Evidence of non-normality:**
    - **Positive/negative skew → asymmetric risk**
    - **High kurtosis → heavy tails**
    - **JB-test → rejects normality (p = 0)**



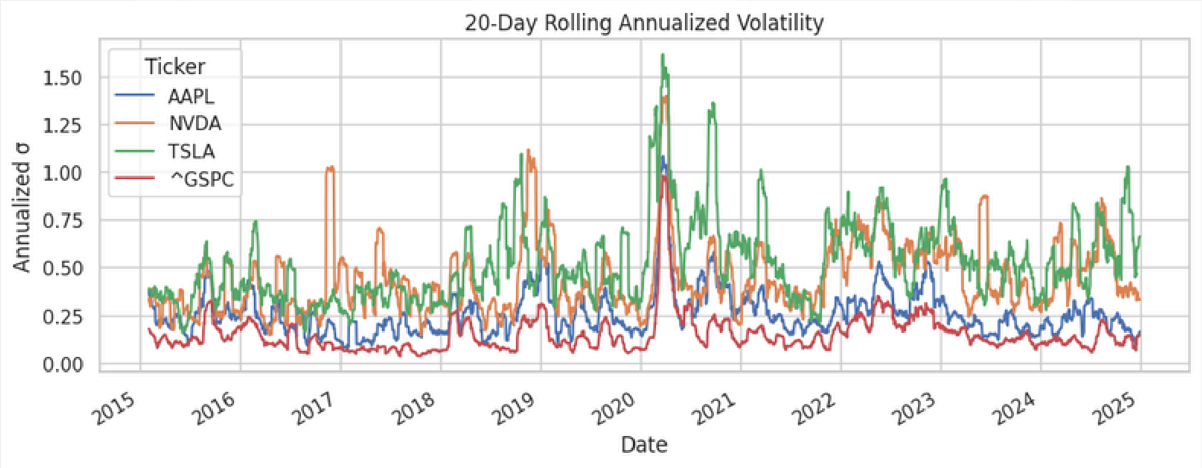
Data Preprocess_Extreme Events (|z| > 3)

Asset	Count
TSLA	40
AAPL	40
NVDA	29
S&P500	36

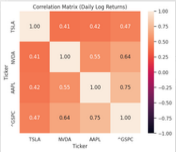
TSLA & AAPL most prone to tail shocks → candidates for volatility domino effects.



Data Preprocess_Rolling Volatility



- TSLA & NVDA show sharp volatility spikes (2020–2021)
- SPX volatility rises after mega-cap spikes → Possible contagion
- AAPL = moderate, smoothed volatility (market stabilizer)



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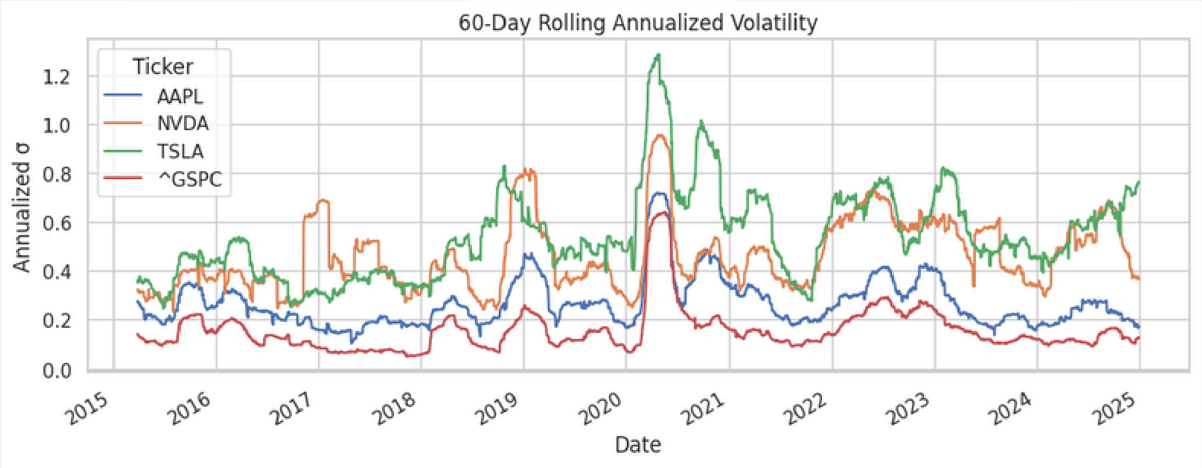

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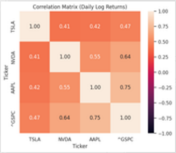

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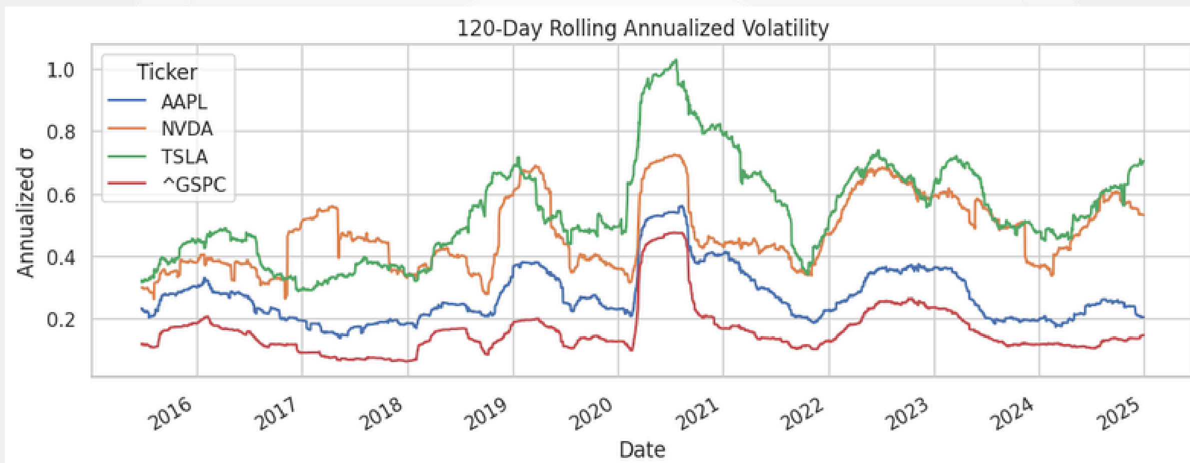

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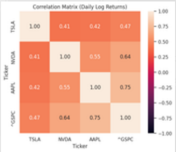
Data Preprocess_Market Sensitivity

Asset	Beta vs S&P500
NVDA	1.73
TSLA	1.5
AAPL	1.19

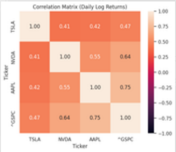
- NVDA & TSLA amplify market risk (“leveraged exposure”)
- High Beta supports volatility transmission hypothesis



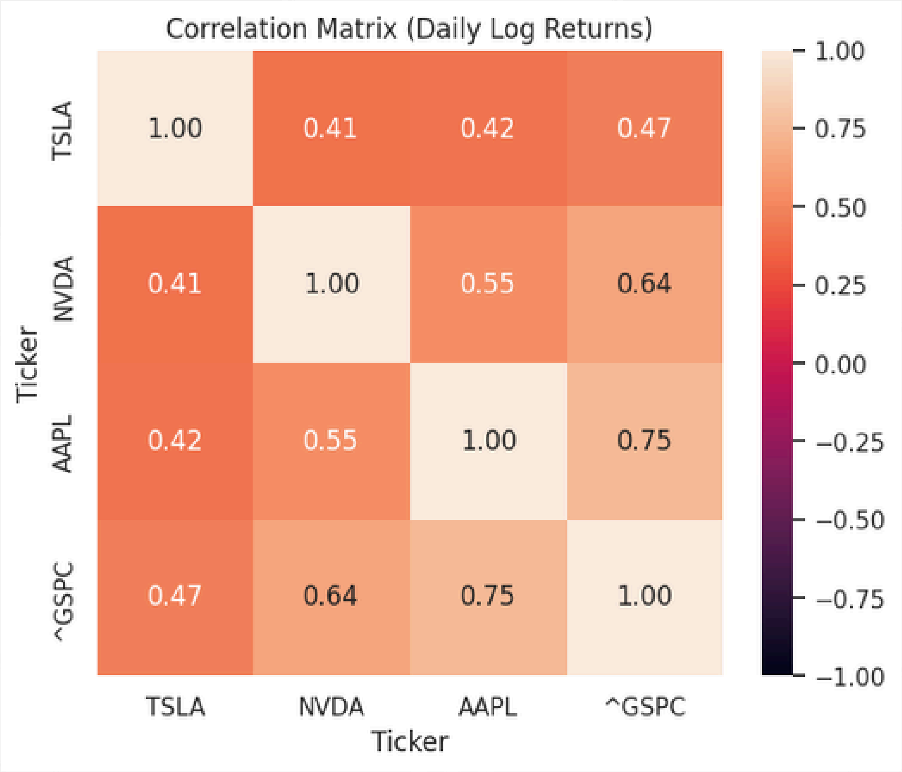
Data Preprocess_Risk Metrics (MaxDD,CVaR)

Asset	MaxDD	CVaR(95%)
TSLA	-73.60%	-0.083
NVDA	-66.30%	-0.067
AAPL	-38.50%	-0.042
S&P500	-33.90%	-0.028

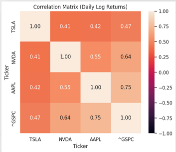
- Mega-caps have deeper crash risk → Potential systemic propagation.



Data Preprocess_Correlation & Co-Movement

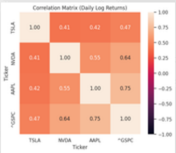


- AAPL highly correlated with SPX (~0.75)
- TSLA shows episodic decoupling (idiosyncratic risk)
- NVDA moderately correlated, rising in AI boom
- Conclusion: Strong integration but not perfect coupling → transmission, not redundancy.



Data Preprocess_Key Findings (Volatility Transmission)

- Evidence of fat tails, extreme risk, and high Beta in mega-caps
- S&P500 volatility spikes lag mega-cap volatility surges
- Tech sector acts as a volatility leader, not follower



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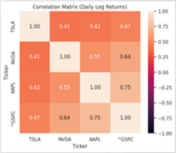


Study Plan

Data Collection

Period
- 2015–2024 (Daily)

Source
- Yahoo Finance (Adjusted Close)

Assets analyzed
- Apple (AAPL)
- Nvidia (NVDA)
- Tesla (TSLA)
- S&P 500 (Benchmark)

DATA PRE-PROCESSING & EDA

Outcome Analysis
- Missing Value
- Logarithm
- Data Completeness
- Outliers Detection
- Variable Transformation
- Multicollinearity
- Special Considerations

Benchmark Model

Benchmark Model
- Linear Regression (OLS)

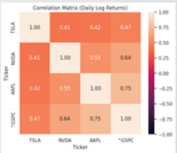
Selected Model

Comparison model
- XGBoost Regression
- Lasso Regession
- Random Forest

KEY METRICS
- **MAE**
- **RMSE**
- **R square**

Comparison & Conclusion

Comparison results between benchmark model and selected model. Provide insights & research contributions.

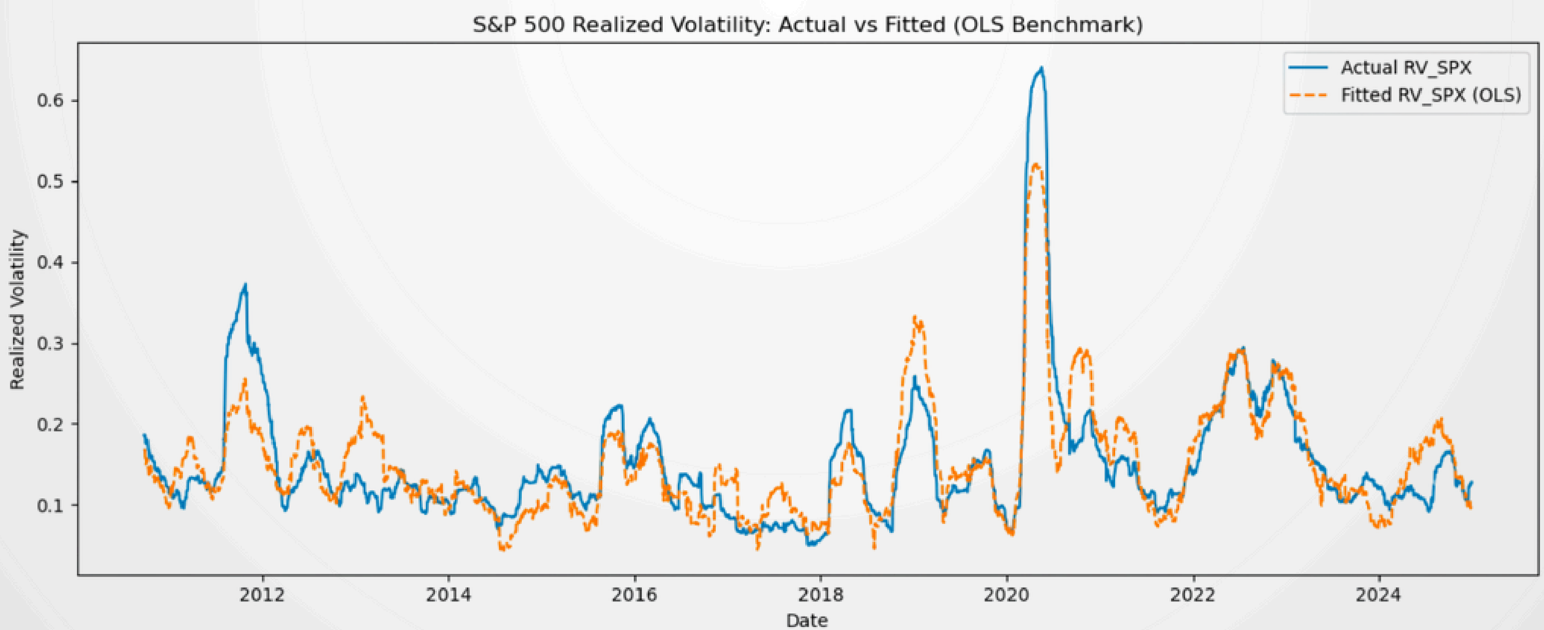


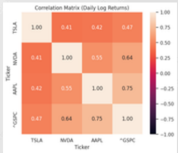
1. Benchmark model: OLS

The baseline regression model

$$RV_t^{SPX} = \alpha + \beta_1 RV_t^{AAPL} + \beta_2 RV_t^{NVDA} + \beta_3 RV_t^{TSLA} + \varepsilon_t,$$

where RV_t^{SPX} denotes the 60-day realized volatility of the S&P 500 and RV_t^{AAPL}, RV_t^{NVDA}, and RV_t^{TSLA} are the corresponding realizations for Apple, Nvidia, and Tesla.





1. Benchmark model: OLS

The baseline regression model

$$RV_t^{SPX} = \alpha + \beta_1 RV_t^{AAPL} + \beta_2 RV_t^{NVDA} + \beta_3 RV_t^{TSLA} + \varepsilon_t,$$

- The model explaining **75%** of total variation.
- **Apple and Nvidia** exhibit the largest and most statistically significant transmission effects.
- **Tesla's effect,** while significant, is markedly smaller

```
================ OLS BENCHMARK RESULTS ================
                            OLS Regression Results
==============================================================================
Dep. Variable:                 RV_SPX   R-squared:                       0.750
Model:                            OLS   Adj. R-squared:                  0.750
Method:                 Least Squares   F-statistic:                     3591.
Date:                Sat, 06 Dec 2025   Prob (F-statistic):               0.00
Time:                        14:04:23   Log-Likelihood:                 6336.3
No. Observations:                3591   AIC:                        -1.266e+04
Df Residuals:                    3587   BIC:                        -1.264e+04
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.0971      0.003    -36.352      0.000      -0.102      -0.092
RV_AAPL        0.5295      0.010     51.261      0.000       0.509       0.550
RV_NVDA        0.1879      0.006     32.301      0.000       0.176       0.199
RV_TSLA        0.0479      0.005      9.076      0.000       0.038       0.058
==============================================================================
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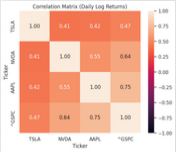
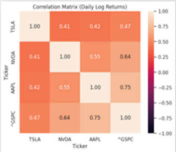
→ Provide clear evidence that a small cluster of mega-cap technology firms serves as **dominant drivers of S&P 500 volatility**



2. Selecting Models

Objective: Exhibit nonlinearity, multicollinearity, and high-dimensional interactions that OLS cannot fully capture

Model	Key Strength	Rationale for Volatility Spillover Analysis
LASSO	Feature selection + multicollinearity control	Identifies core spillover transmitters; **stabilizes coefficients** among highly correlated mega-caps
Random Forest	Nonlinear modeling + interaction capture	Detects **nonlinear volatility spillovers** and structural breaks in tech stocks
XGBoost	Highest predictive performance in finance	Learns complex **crisis-sensitive** and **asymmetric transmission patterns**; handles outliers well



2. Selecting Models

XG BOOST

The XGBoost prediction is

$$\hat{y}_t = \sum_{m=1}^{M} f_m(\boldsymbol{x}_t) = \sum_{m=1}^{M} f_m\left(RV_t^{AAPL}, RV_t^{NVDA}, RV_t^{TSLA}\right).$$

The trees $\{f_m\}$ are obtained by minimizing the regularized objective

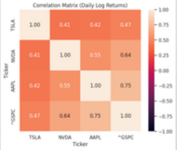
$$\mathcal{L} = \sum_{t=1}^{n} \ell(y_t, \hat{y}_t) + \sum_{m=1}^{M} \Omega(f_m),$$

where $\ell(\cdot)$ is typically the squared loss

$\ell(y_t, \hat{y}_t) = \frac{1}{2}(y_t - \hat{y}_t)^2$,

and $\Omega(f_m)$ is the tree regularization term (penalizing number of leaves and leaf weights)

Model	RMSE	MAE	R^2
OLS (Linear Regression)	0.032018	0.025008	0.709753
XGBoost	0.041913	0.032884	0.502621



2. Selecting Models

Random Forest

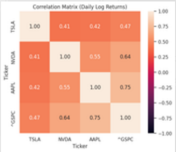
Let $T_b(\boldsymbol{x}_t)$ be the prediction from the b-th tree ($b = 1, \ldots, B$).

The Random Forest prediction for day t is

$$\hat{y}_t = \frac{1}{B}\sum_{b=1}^{B} T_b(\boldsymbol{x}_t) = \frac{1}{B}\sum_{b=1}^{B} T_b\left(RV_t^{AAPL}, RV_t^{NVDA}, RV_t^{TSLA}\right),$$

where each T_b is trained on a bootstrap sample of the data and at each split only a random subset of predictors is considered

Model	RMSE	MAE	R^2
OLS (Linear Regression)	0.032018	0.025008	0.709753
Random Forest	0.048793	0.037036	0.325929



2. Selecting Models

LASSO

Prediction has the **same linear form** as OLS:

$$\hat{y}_t = \hat{\alpha} + \hat{\beta}_1 RV_t^{AAPL} + \hat{\beta}_2 RV_t^{NVDA} + \hat{\beta}_3 RV_t^{TSLA}.$$

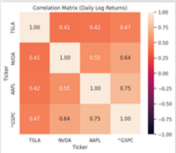
The difference is **how** $(\hat{\alpha}, \hat{\boldsymbol{\beta}})$ are estimated.

LASSO solves the penalized least-squares problem (what you did with `LassoCV` in Python):

$$(\hat{\alpha}, \hat{\boldsymbol{\beta}}) = \arg\min_{\alpha, \boldsymbol{\beta}} \left\{ \frac{1}{2n} \sum_{t=1}^{n} \left(y_t - \alpha - \boldsymbol{\beta}^\top \boldsymbol{x}_t \right)^2 + \lambda \sum_{j=1}^{3} |\beta_j| \right\},$$

where $\lambda \geq 0$ is the regularization parameter chosen by cross-validation.

Model	RMSE	MAE	R^2
OLS (Linear Regression)	0.032018	0.025008	0.709753
LASSO	0.031233	0.025042	0.723805 ✔



Machine-learning Model Performance

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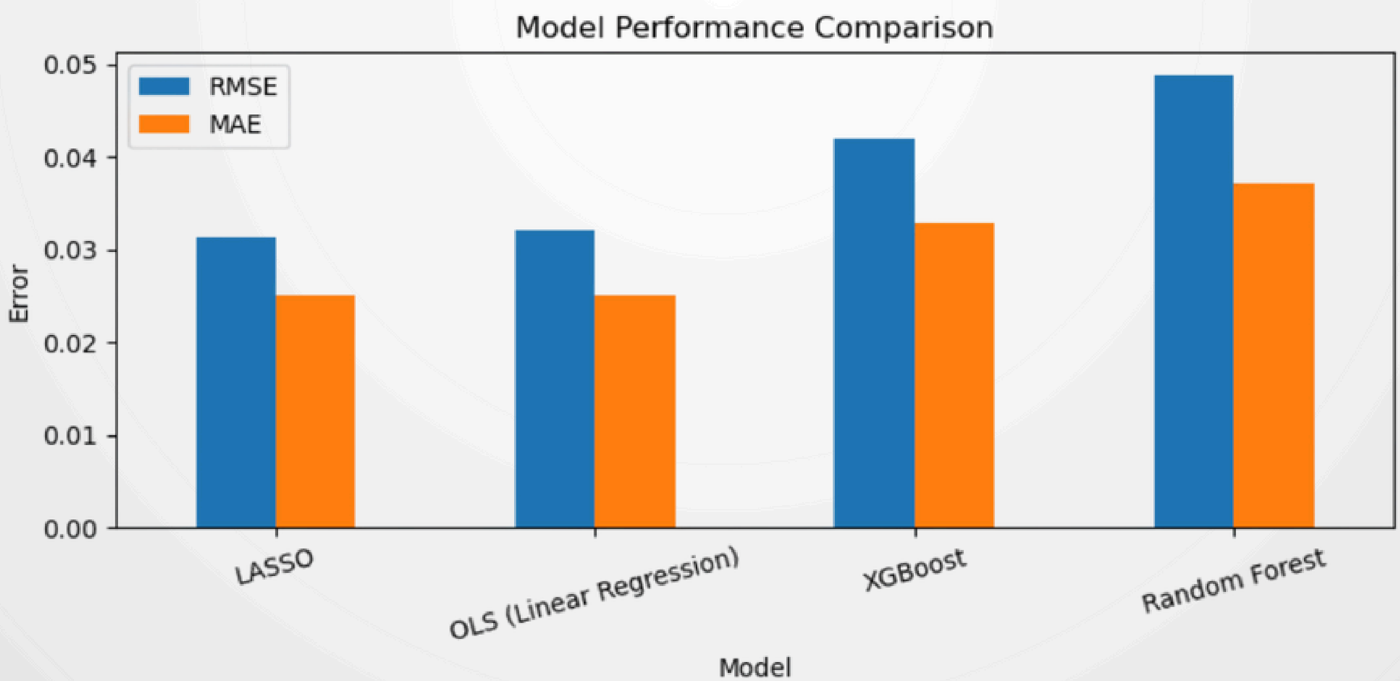
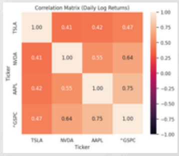
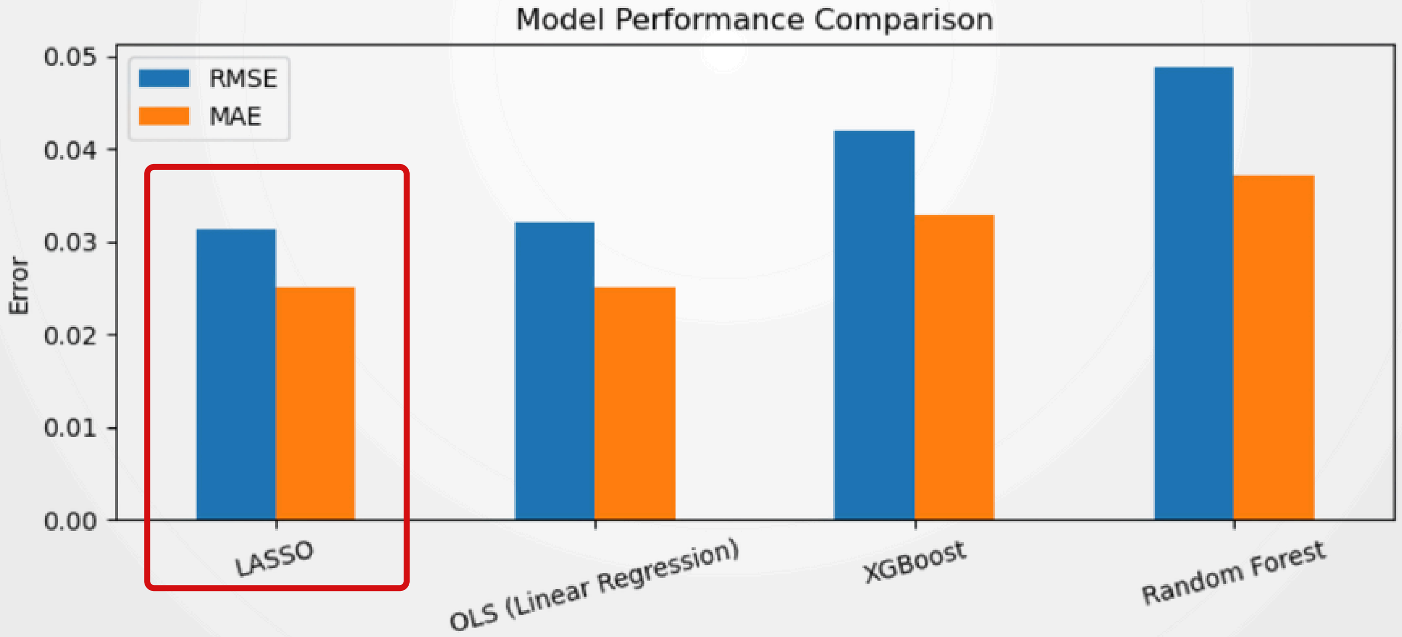


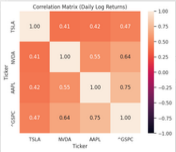


Machine-learning Model Performance

LASSO achieves a better bias–variance tradeoff

- A lower RMSE → Better predictive accuracy
- A higher R square → more variance in S&P 500 volatility.
- Comparable MAE to OLS → no loss in average prediction error

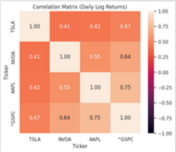




Key insight & Contribution

- **Apple**’s realized volatility exhibits the **strongest spillover effect.**
- Followed by **Nvidia**, while **Tesla**’s influence is positive but **comparatively modest**, indicating that its volatility is more idiosyncratic.

Small set of large technology firms plays a disproportionately systemic role in shaping aggregate U.S. market risk



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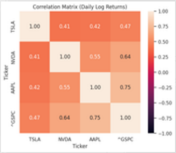
Correlation Matrix (Daily Log Returns)
TSLA
NVDA
AAPL
^GSPC
Ticker
1.00
0.41
0.42
0.47
0.41
1.00
0.55
0.64
0.42
0.55
1.00
0.75
0.47
0.64
0.75
1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00